Hum Ecol (2007) 35:69–80
DOI 10.1007/s10745-006-9082-5

Making Sense in Time: Remote Sensing and the Challenges of Temporal Heterogeneity in Social Analysis of Environmental Change—cases from Malawi

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Published online: 12 December 2006

Abstract This article contributes to the encouraging recent trend toward more critical examination of the linkages between remote sensing and social analysis of environmental change. Using two case studies from Malawi, we explain the importance of thinking about temporal scale when using remote sensing data. Specifically, from our on-the-ground research, we came to appreciate three important issues: (1) multiple causation and multiple temporal scales operating simultaneously—making inference of casual relations between particular social factors and observed land use changes difficult; (2) 'visible' vs. 'invisible' processes and events; and (3) 'continuous' vs. 'punctuated' social and environmental processes. We observe that in each case, social differentiation is key to understanding which temporal frames are most relevant in understanding observed land use dynamics. We conclude that, while these are not entirely new observations, research on the applications of remote sensing in social analysis of environmental change could be enriched by more rigorous examination of linkages between environmental change, temporal scale, and the social relations (including social differentiation) that can help to explain how and why particular temporal frames are most salient.

Key words Malawi · temporal heterogeneity · remote sensing

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Introduction

This article uses two case studies from Malawi to examine questions about the integration of remote sensing imagery into social research on environmental change, with specific focus on the role of the temporal scales of social dynamics, and with attention to the role of social differentiation. As a geographer and a social anthropologist, we collaborated in field studies to address the complex interplay of agro-ecology and sociopolitical history in one of Africa's most densely populated regions. While sheer numbers are clearly relevant to understanding land use change, at the center of our concerns were obvious and not-so-obvious national and localized social differentiation in the rural sector, and their implications for land use as policies and practices shifted over the last decades of the twentieth century. We found that differentiation was relevant to understanding both long-term processes and fairly rapid punctuated shifts but in ways that required both methods to fully explore. Social gradations that affect livelihoods within the smallholder sector can escape synopses based on remote sensing, and the magnitudes and speeds of change can escape fieldwork. By comparing two areas over the same period of political and economic history we were able to clarify salient relationships between the stratification of the rural sector in Malawi and land use dynamics.

Recent years have seen a very productive expansion of critical research on the challenges of using remote sensing and geographic information systems to understand environmental change from a social perspective (Craig *et al.*, 2002; DeSanker *et al.*, 2005; Fox, 2003; Geoghegan *et al.*, 2001; Geoghegan and Gray, 2005; Heasley, 2003; Jiang, 2003; Robbins, 2003; Sheppard and McMaster, 2004; Turner 2003; Turner and Taylor, 2003; Turner *et al.*, 2004; Vance and Geoghegan, 2002; Zubrow, 2003). This expanding literature generally focuses on social constructions, multiple interpretations, and, in some cases, contestations of previously unproblematized categorizations of environmental data as viewed through remote sensing. Most of these critical assessments focus on cultural and political-economic heterogeneities that produce differing views and interests in the meanings of environmental change data. Although questions of temporal heterogeneity are implied in some of these discussions, relatively few examine temporal issues explicitly, and especially with respect to variables beyond population change and macro-political economic forces.

In this article, we contribute to the expanding breadth and depth of research on the use of remote sensing in social research by making some of these implicit questions of time and temporal framing more explicit. Drawing from our research in Malawi, we offer three broad observations. First, landscape change—especially in densely populated, highly commercialized and deeply politicized contexts—is characterized by multiple social and environmental changes occurring *simultaneously, at differing temporal scales*. This makes the choice of particular time frames of analysis critical in shaping the resulting interpretations (see also Sheppard and McMaster, 2004). Second, we suggest that the strength of remote sensing (RS) and geographical information systems (GIS) analysis in detecting processes that are 'visible' with current technologies is inherently selective in that some processes and events occur on time scales and in ways that are readily detectable with RS/GIS technology, while others may be less so. Third, we observe that a particularly important example of the problem of 'visible' and 'invisible' temporal scales in RS/GIS involves the question of how 'punctuated' historical events are assessed in relation to continuous, gradual processes. These questions of scale are, we suggest, often linked to issues of social difference. For example, the relevant scale for examination of such issues as deforestation may be very different for different social groups in the same time and place. We use two case studies from Malawi to illustrate how these questions arose in our research on social and environmental change (specifically, deforestation). We do not suggest that these are by any means an exhaustive list of potential issues of interest or concern regarding the role of temporal scale in research on social and environmental change integrating RS/GIS.

This article is based on research conducted separately and together by the authors since the 1980s and 1990s. Walker is a human geographer and Peters is a cultural anthropologist. Peters has worked extensively in Malawi, where since 1986 she has been conducting fieldwork focused on questions of how broad economic policy shifts (e.g., structural adjustment and food security policies) mediate and are mediated by social context and change at the community, village, and household scale. Walker has worked periodically in Malawi since 1990, including fieldwork in 1995–1996 on social and political dynamics affecting reforestation efforts in densely populated and heavily deforested areas. We are not expert practitioners of remote sensing by any stretch of the imagination. However, we have used time-series aerial photography (and to a very limited extent satellite imagery) enough to be aware of some of the opportunities and challenges that these technologies pose for social scientists studying social and environmental change. This article is intended to contribute to the literature on the use of remote sensing in social analysis in part by recounting some of the questions and problems encountered by *non*-RS/GIS specialists.

The Setting: Changes in an Agrarian Society

Our discussion is based on fieldwork in villages in the densely populated Zomba District in the Shire Highlands of southern Malawi, and the less densely populated Kasungu District of Malawi's central region. Peters selected the Zomba villages in 1986 as representative of areas experiencing population/land pressure as well as commercialization of agricultural production. In 1995–1996, Walker added the Kasungu site for his dissertation fieldwork on the social conditions affecting reforestation programs because it is broadly comparable to the Zomba site in cultural and economic terms, but has experienced relatively less population/land pressure. Before discussing some questions about RS/GIS and temporal scale in the following sections, we provide background on salient changes in these regions—primarily demographic changes and changes in economic activities. These range from ongoing processes that date from the late nineteenth century to more recent changes since the 1980s and 1990s.

Demographic Change

As a whole, Malawi's demographic history during the late nineteenth century, the twentieth century, and (by all signs) continuing into the twenty-first century has been characterized by very rapid population growth. In the year 1890, the population of the then-colonial state of Nyasaland as a whole was recorded at 543,000 (Utrecht University Library, 2005a, 2005b); the 1998 national census of Malawi recorded 9.9 million. The present overall national population density is estimated at 127 persons per square kilometer, one of the highest in Africa (Benson *et al.*, 2002). The history of population growth has been phased and scaled differently in different regions, as we discuss below, but for the period for which reliable statistics are available the variation has all been at the high end of rates recorded in Africa. From 1945 to 1990, the population of Zomba District grew by 225%, while the population of Kasungu District grew by 770% (Utrecht University Library, 2005a, 2005b). Between 1977 and 1998, Malawi's population grew at an average annual rate of 2.77%, while the population of Zomba District grew at a rate of between 1.5% and 1.98% per year, and the population of Kasungu District grew at between 3.5% and 5.0% per year (Benson *et al.*, 2002).

This dramatic growth must be understood in historical context. Around the middle of the nineteenth century, the plateau area of the Shire Highlands was largely emptied by a combination of slavery, conflict between Nyanja and Mang'anja people of the Highlands and nearby Shire with the Yao pushing in from what was becoming Portuguese East Africa (Mozambique), and associated famines and devastation. Later in the nineteenth century and into the twentieth century, the peace imposed by British rule was accompanied by people moving back down from the hill fortresses onto lower land and the appropriation ("purchase") of land, much of which had previously been nearly emptied by slavery and warfare, for European tobacco and tea estates. Having failed to squeeze enough labor from the existing populations on estate land (who were declared "tenants" and forced to work for their "rent"), the estates welcomed families fleeing from the harsh conditions of Portuguese rule across the newly won border. The 1920s and 1930s thus saw a huge and rapid influx of migrant households in addition to those returning to their ancestral land after the social disruptions of the late nineteenth century (Vail, 1983; White, 1985). This, in conjunction with natural population growth, created intense pressure on both private estate land and surrounding "crown" (later "customary") land. The stage was set for many decades of tense negotiations and even violence (Mwase and Rotberg, 1967) in the Shire Highlands between the sparsely-populated European estates and Africans living in adjacent overcrowded customary areas—tensions over land, labor, and access rights that continue in new forms even today (Walker and Peters, 2001).

Farther from the violence, and exodus from Portuguese East Africa, the Kasungu District of the central region did not experience the same scale of in-migration in the late nineteenth and early twentieth centuries. But from the 1920s onward Kasungu District became an important destination for small farmers fleeing overcrowding in the south, as well as newly emerging African elites seeking to establish their own estates. In the 1970s and 1980s especially, Kasungu District became the chief area of land appropriation by the government of "Life President" Kamuzu Banda for himself and others connected to the ruling party (Dickerman and Bloch, 1991). The population density in Kasungu District remains among the lowest in the country, but the district experienced remarkably rapid population growth in the latter half of the twentieth century as new estates attracted additional workers, and as migrants from other, more crowded, areas migrated into the district's still relatively land-abundant customary areas. The rate of growth in Kasungu in the late twentieth century was comparable to the rapid growth in the south a half-century earlier.

Economic Change

With this growing land pressure from in-migration, natural population growth, and appropriation of land for private estates, small farmers in both of research areas attempted various strategies to diversify and intensify their economic activities. In the early twentieth century, these efforts were often blocked by colonial authorities whose first political priority was to protect European estates from competition for labor and commercial markets. Many have argued that the colonial government of Nyasaland created one of Africa's most ruthlessly exploitative systems of neglect and 'taxation' of small farmers for the benefit of the estate sector (Mhone, 1987). In addition, as a landlocked nation with poor transportation infrastructure and (unlike nearby South Africa and Rhodesia) little mineral wealth, Nyasaland struggled to find its economic footing. By the period of independence in the late 1950s and early 1960s, even the European tobacco estates were generally not profitable. Nyasaland remained a poor and forgotten colonial backwater, a land Vail (1975) described as an "imperial slum." Malawi emerged as an independent nation after 73 years under colonial rule as one of the poorest countries in Africa, where a sizeable proportion of rural Malawians are chronically unable to feed themselves during each year's preharvest 'hunger' season, and where land pressure and lack of investment capital appear to foreclose any clear path toward progress for many rural people.

For a brief period, independence from Britain in 1964 brought a glimmer of hope. The independent Government of Malawi broke up some of the former colonial estates in the southern region and converted them into customary land so that the residents were freed of the hated tenant labor system, locally referred to as *thangata*. This period of increased access to land, removal of tied labor, and relaxation of legal restrictions on crops that could be grown and sold by ordinary Malawians led to what seems to have been expansion of agriculture and an increase in overall welfare during the late 1960s and early 1970s. But the improvement, as well as hope for a drastically different type of government or economy, proved short-lived. The president, Dr. Hastings Kamuzu Banda, actually intensified (under new names) the old systems of 'taxation' and restrictions on small farmer agriculture, while channeling capital to the estates and appropriating land for himself and his political supporters (Kydd and Christiansen, 1982). During the 1970s and early 1980s he expanded estate land enormously, although most of the new estates were smaller and more fragmented than the enormous European ones. Because many of these new estates were carved out of customary land (in many cases displacing whole villages), they acted to reduce availability of land to small farmers in some areas. With Banda's renewed land appropriation, by 1993 the area controlled by estates had grown to 173% of the peak level under colonial rule (United Nations, 1993). By the mid-1980s, land appropriation and natural population increase had led to permanent cultivation of land by smallholders in much of the crowded southern region. Soil fertility had declined across most of the country and, in the most crowded areas, increasing cultivation on steep slopes and in wetlands, as well as increased demands on seasonal water sources, represented significant problems for much of the population. As we discuss in the following sections, one of the important responses by small farmers to these intense pressures has been efforts to diversify crop production, although both the pattern and scale of these efforts remain poorly understood. In our research sites, almost all villagers own (under customary tenure) at least a small amount of land, and prioritize working this land for their own food and cash crops. However, many, especially those with the least land, supplement their income by working part-time for other small farmers and local estates, engaging in petty trade or small-scale services, or in low-wage work in neighboring towns. Many families in the Zomba sample have adult children employed full-time in towns, some of whom send remittances but who also often receive gifts (usually of food) from their rural homes. Despite these highly diversified income strategies, many remain poor. Research over the period 1986–1997 in Zomba also revealed an increasing disparity in income levels between the top income quartile and the bottom income quartile.

Such livelihood problems have been less pronounced in the central region and Kasungu District, but land pressure is becoming a problem there as well. Kasungu District is now the center of much of the nation's tobacco production. Land in this area was relatively abundant throughout the colonial period, and much of the area remained heavily wooded. From the 1970s, however, appropriation of land for large leasehold tobacco estates (typically leased by government to private individuals as political favors), along with natural population growth and immigration from other areas, contributed to significant changes. From the 1920s, attractive wages in the mines of South Africa induced large-scale migration of male labor out of Kasungu District, reducing the area that could be cultivated in the home villages and contributing to regrowth of large areas of woodlands (McCracken, 1987). With the closing of opportunities for labor migration to South Africa in the 1980s, more male labor was available with a consequent increase in the area and intensity of cultivation. The scarcity of wage opportunities encouraged some households to increase their production of cash crops (tobacco, soy beans, and sunflower seeds) while maintaining previous levels of food crop production, thus requiring the clearing of land previously left fallow. However, the ability to expand production onto new land was limited primarily to the earliest settlers in the area, whose families controlled enough land to meet their subsistence needs and leave some fallow. Relative latecomers secured much smaller landholdings, which constrained their ability to expand production into cash crops. While growing land pressure represented long-term settlement trends, the expansion of cash cropping in the 1980s partly represented a sudden, punctuated increase in land pressure.

Not all changes in the smallholder economy were in the nature of defensive moves to preserve eroded livelihoods, however. In both the Shire Highlands of the south and Kasungu District of the central region, one of the most important economic changes (with possibly important ecological consequences) was the removal in the early 1990s, under policies of 'structural adjustment' imposed by international financial institutions, of restrictions on small farmer production of burley tobacco. In brief, our research shows that the effects of burley production have been both good and bad (as well as ambiguous) for smallholder incomes (cf. Orr and Mwale, 2001). The good effects include a sharp rise in income for the bigger smallholder producers (a relative term, since many of even the "big" smallholder producers usually cultivate less than half a hectare of tobacco). By the mid-1990s, the improvements in material goods (housing, clothes, bicycles, etc.) as well as food supplies in the better-off families were clear. The negatives may include the effect on forests (as discussed later in this article), the increased use of pesticide in tobacco nurseries along river beds (see below), and the effects on soil where the tobacco is intercropped or rotated with incompatible crops leading to a build-up of nematodes in the soil (Peters, 2002). In Kasungu District, almost all smallholders who engage in burley production do so by expanding their cultivated area into land that is otherwise uncultivated. Only the smallest landholders (almost all in our southern research area) find that burley production reduces food supplies when they reallocate land from food production to burley, which—when it fails—leaves them without income and without food.[1]

[1] Current research in Zomba (2006) shows that the recent low prices received for burley tobacco through the Auction Floors in relation to the increase in costs of fertilizer and credit charges are leading many to reduce or abandon tobacco growing. This may again reverse if these price/profit differentials shift.

The other type of land use that is increasing, at least in the southern region, is riverbank and riverbed gardens, referred to as *dimba*. In Peters' early research (1986–1987), it became clear how important *dimba* were to people's livelihoods and also to increasing competition over land. In a climate with a single rainy season and unpredictable rainfall patterns, and with the growing demand for farmland, dry season cultivation in *dimba* has become even more important. Furthermore, the increase in burley production has led to an increased need for *dimba* for tobacco-seedling nurseries. Growth in building in urban areas and rural market centers has stimulated demand for river sand and gravel. In addition, recent attention to irrigation by government and donors has intensified interest in watered land. These demands all converge on rivers, riverbanks, and increasingly in wetlands (Kambewa, 2006).

There have been several other recent and important changes in this agrarian society that are relevant to our discussion of the role of remote sensing in social analysis. First, Malawi is one of the many African countries that shifted toward a more democratic form of governance in the mid-1990s. In 1994, the country held its first multiparty elections, and has since held elections in 1999 and 2004. The question of how truly 'democratic' these elections were is a matter of ongoing debate, but the consequences of this 'opening up' of Malawian society were dramatic—not least for the physical environment (see Walker, 1999). Another, more frightening, turn of events has been the recent catastrophic explosion of the HIV/AIDS pandemic. Recent studies indicate that the loss of labor due to AIDS-related illness and deaths and associated declines in labor-intensive crop diversification, particularly at the critical planting and harvesting seasons, as well as loss of crucial agricultural knowledge, contributes to declines of nearly 50% of household food production in the most severely affected regions of Africa (United Nations Department of Economic and Social Affairs/Population Division, 2003, p.62). The full impacts of this catastrophic disease on smallholder households, agriculture, and the environment in Malawi remain poorly understood and are currently a top research priority. Particularly important is research that, as in Peters' current research, is able to draw on the history sketched in order to compare accounts of life before the epidemic with the present.

We have briefly sketched only a few of the important changes in the rural society and landscape of Malawi since the late nineteenth century, but this provides sufficient parameters for our study of the social and environmental changes at a finer scale in the two case studies. As outlined earlier, these two villages are broadly indicative of conditions in Zomba and Kasungu Districts. We focus particular attention on changes in tree cover and the social forces that may drive these changes. Tree cover is one of the clearest indicators in remote sensing, and so offers strong evidence. It is also subject to several different pressures, ranging from commercial exploitation for charcoal or building, clearing for new fields in farming systems where an extensive solution is still viable, provision of domestic fuel for the next generation of farmers, and clearing of forests for intensive cultivation, amongst others. The location, kind, degree, and, above all, timing of change in tree cover may still be identifiable as responsive to different social dynamics, even in areas where a great deal is going on at once. In fact, the strength of combining methods should be the ability to tease out and interrelate these different social and environmental dynamics on their own temporal scales.

Social and Environmental Change in Two Villages

Zomba: Village Use of Trees on Private Estate Land

Southern Malawi is often described as a place of environmental crisis—with very high population density and among the highest rates of deforestation in Africa, and with very high rates of soil erosion and depletion. Significantly, the language of 'crisis' has been present in southern Malawi for at least 70 years, with archival records indicating high levels of environmental concern among colonial administrators from the 1930s onward. This has continued to the present, most notably in Malawi's 1994 National Environmental Action Plan. This case study, based on research conducted in Zomba District between 1995 and 1997, examines the response of small farmers to government initiatives to encourage tree conservation and tree planting (see Walker, 2004).

As discussed in the previous section, the creation in the early colonial period of the estates on former village land and the later redistribution of vacated estate land to nearby villagers and resettled migrants during the period of independence in the 1950s created a mosaic of land ownership, with many small and fragmented estates interspersed between areas of customary land. Although our research sites are not necessarily representative of all areas of the country, this fragmentation of land ownership is easily visible in remotely sensed images. Even old and relatively coarse-scale (1:40,000) black and white aerial photographs show sharp, clear lines that correspond with known boundaries between government, estate, and customary land. In our Zomba site, tree cover sharply differs between customary and estate land. Customary land is virtually empty of closed-canopy tree cover. The only significant tree cover consists of individual fruit trees and village cemeteries. Estate land varies, but generally has more tree cover, including both planted trees (usually eucalyptus woodlots for poles and firewood) and naturally-regenerated indigenous hardwoods on uncultivated land (some estates leave land uncultivated due to insufficient capital).

Using combined ethnographic and survey methods, 89 households were interviewed about their use of trees and tree products. When asked how they obtain vital wood products (firewood, building materials), many respondents openly state that "we steal" (*timakuba*) trees from neighboring estate land—creating an ongoing source of conflict between villagers and private estate owners (Walker and Peters, 2001). Notably, claims to rights of access to trees and tree products on estate land are based upon established historical claims, and these claims are strongest among long-time residents who were present during the period of European rule, especially those who claim their family land was taken by the Europeans and those whose families once worked for the colonial estates. To some extent younger generations and migrants appear to 'free-ride' on these historical claims, and the durability of these claims as the older generation passes away remains an interesting and unexplored research question. Nevertheless, at present the key observation is that in contrast to the views of the government, most peasant farmers do not (yet) perceive tree scarcity as rising to the level of a 'crisis' because most have access (albeit contested) to trees on private land they do not own. Most smallholders perceive tree scarcity as primarily a problem of *access* to trees rather than a problem of absolute scarcity.

By comparing aerial photography from 1965 and 1995 it is clear, however, that deforestation is in fact occurring on a large geographic scale even while local pockets of protected forests on private land remain. While much of the focus of attention has been on the large-scale loss of forest cover that is readily visible from remotely sensed images, our research suggests that these local 'pockets' of protected forest are critically important sources of material resources (firewood, poles, lumber) for large numbers of people, and could be easily overlooked.

The importance of considering differences in geographic scale in shaping perceptions of tree scarcity is vividly illustrated in particular by the case of the Madsen Estate[2] on the Thondwe River. Aerial photography shows that since 1965 there has actually been substantial regrowth of indigenous hardwood forests (*miombo*) on the Madsen Estate despite the broader pattern of deforestation in the region. Local respondents explain that until his death in the late 1980s, Mr. Madsen (a mixed-race Malawian) was particularly vigilant in patrolling his estate (reportedly with a gun) against 'poachers' (i.e., neighboring villagers). While forests on some other estates became exhausted, the Madsen Estate retained its forests intact. On his death, Mr. Madsen's disabled widow was unable to effectively protect against poaching, and the estate became the new de facto woodlot for nearby villagers. Such small but critical spaces have now become key sources of wood products. The legal institutions to enforce private property rights have been key to overall land use and to the livelihoods of the poor (e.g., Thompson, 1975, on poaching), including the environmental effects of social stratification. This suggests the need for a grounded and institutional understanding of social differentiation.

The question of what kind of environmental 'crisis' exists for villagers depends on both the geographic and temporal scale that one chooses to examine. The 1965 and 1995 aerial photographs show increased land pressure from an already high level, with cultivated area within the village increasing from 91% to 98%, leaving virtually no uncultivated land except in culturally-protected cemeteries. However, the question of the impact of tree use is difficult to measure because the village appears to rely heavily on tree resources outside village boundaries, and the locations from which villagers acquire tree products has shifted over time from one neighboring estate to another. For example, villagers in our research site report that before Mr. Madsen died, they relied heavily on trees 'poached' from the Magunda Estate. Aerial photographs show that from the 1960s to the 1990s, tree cover on the Magunda Estate dramatically decreased (although it still has significant areas of planted eucalyptus). Villagers reported that with the death of Mr. Madsen in the 1980s, their primary source of tree products shifted from the Magunda Estate to the Madsen Estate, reflecting both the increased scarcity of trees at Magunda and the 'opening' of access to trees on the Madsen Estate. By the 1990s a few villagers still openly acknowledged that they 'steal' trees from the Magunda Estate (and Mr. Magunda still employed guards to protect his planted eucalyptus). However, oral accounts overwhelmingly suggest that the primary source of trees for most villagers shifted to the Madsen Estate. Thus, while a comparison of standard 9×9-in. aerial photographs that cover approximately 8,400 ha, gives the overwhelming first impression of tree cover loss from the 1960s to the 1990s, to the local people, whose villages typically cover no more than a few hundred hectares, a geographic frame of 8,400 ha is not a particularly relevant scale. To these villagers, perceptions of tree 'scarcity' appear to be defined more in terms of the availability of small 'pockets' of trees that could easily be overlooked in an aerial photograph, and by the *conditions of access* to these resources, which are, of course, entirely invisible in aerial photographs. Similarly, local perceptions of whether trees are becoming more or less scarce over time is a matter of fine-scale geographic and social conditions not easily perceived in simple before-

[2] All local names of people and places in this article are pseudonyms.

and-after comparisons of aerial photographs. This helps to explain why villagers reported to us, to our surprise, that they did not generally perceive tree scarcity as a growing 'crisis'—a response that we surmise might have been different had the specific events and conditions on neighboring estates not been what they were. Simply put, we believe our 'surprise' as researchers can be explained in part because the questions we posed and answers we expected were premised on geographic and temporal frames that differed from those of our respondents, and were not especially relevant to them.

Kasungu: Tree 'Poaching' on National Park Land

By the 1920s, Kasungu District was at the center of expansion by European tobacco estates, and the relative abundance of land also allowed for the creation, in 1922, of the Kasungu National Park,[3] which today remains the nation's largest. With land occupied by older European estates, the National Park, and the many smaller estates created in the postcolonial period, the once sparsely populated Kasungu District became, like the more densely populated south, a mosaic of land ownership in which customary land is squeezed between state and private property. Of particular importance is the highly differentiated pattern of land ownership within villages, where the largest landholders today are the descendants of the pioneer settlers who came to the area in the early twentieth century, and those with the smallest landholdings are migrants, many of whom were driven off their land since the 1980s by the creation of new estates in other areas.

Our research village in Kasungu is situated on approximately 4 km^2 of land bordered on three sides by relatively heavily populated smallholder villages and by a large forested area of the Kasungu National Park on other. The present headman and others date the first settlement of the village to 1918. At the time, the entire area was 'bush,' and villagers were reportedly attacked by lions. In 1920 the village relocated just across the border in Zambia. In 1933, after the park was established and the increased human population had driven away the lions, the original village headman moved his people back to Kasungu. But the village had a new neighbor—the 231,600-ha Kasungu National Park. People in the village today claim that the park took part of the land they originally settled in 1918, and that the government should give it back. While the exact boundaries of the original village settlement and the degree to which they overlapped with the boundaries of the national park remain vague, this oral history and a sense of moral entitlement strongly infuse resource politics in the area up to the present.

These resource politics have become increasingly acute as forests, woodlands, and long-term fallow have decreased. Unlike the Zomba site, 1962 aerial photographs show that much of the village land remained forested at the time, with only 38% of the total area cultivated. Although the relatively poor resolution and quality of the 1962 photographs makes accurate classification difficult, they appear to indicate roughly equal areas of cultivated land, long-term fallow, and dense woodlands or gallery forest (especially near wetlands and stream banks, and between cultivated fields). By 1995, 71% of land in the village was under cultivation. However, the relative abundance of forest, woodlands, and long-term fallow in the village—29%, compared to virtually zero in Zomba—has not resulted in an entirely different sense of tree scarcity or abundance. The village has widely differentiated land ownership: on average, the top 10% of landowners (mainly, those whose families settled first in the area) control nearly 20 ha of land per household, compared to 0.61 ha controlled by the 10% of villagers with the smallest holdings (typically those who settled recently after most arable land had already been distributed). Thus, social differentiation of landholdings within the village, coupled with the inability to expand landholdings because of the presence of the national park and surrounding private estates, has effectively created a 'scarcity' of land that is similar to the pattern in the Zomba site.

The differentiated pattern of landholdings in our Kasungu research area has had a dramatic impact on tree use and village relations with the adjacent Kasungu National Park. For this research, 65 households were interviewed about where they obtain trees and forest products. Just as villagers in Zomba declare their right to take tree products from neighboring estate land, the great majority in the Kasungu site almost defiantly state that they 'steal' from the National Park. For example, 71% of households identify the National Park as their primary source of poles for construction, and 58% identify the park as their primary source of firewood. Aerial photographs taken in the early and mid-1990s show a razor-sharp line between the intensively cultivated village land (with patches of fallow and forest) and the heavily forested national park. However, close inspection also indicates a clear pattern of tree thinning along the boundary of the village. Some villagers do not hesitate to reveal that they will cut whole live trees across the park boundary if necessary (mostly for lumber for construction or sale).

However, the aerial photographs also indicate that clear-cutting of large swaths for forest within the park does not

[3] The park was first declared a forest reserve in 1922, became game reserve in 1930, and was finally declared a national park in 1970. The land now occupied by the park was inhabited in 1922, but the relative abundance of land in the region at the time made it relatively easy to relocate occupants to nearby areas. This would have been much more difficult in the densely populated south.

occur (almost certainly because of fears of retaliation by park rangers). Many villagers voiced a desire to cultivate or graze land within the boundaries of the park, but none reported actually doing so, and aerial photographs show no clear signs of such activities. Quite horrific stories—entirely unverifiable—of abuses by park rangers of villagers caught 'stealing' in the park appear to be the main reason that people are limited in their use of park land.

However, at the time of these interviews (1995–1997), a new expanded rhetoric of entitlements had emerged. In the 1994 presidential and parliamentary election, candidates had freely wielded promises to redistribute land in the National Parks and forest reserves to those in need—promises that were wildly popular, and appeared to become the leading edge of a reworked sense of moral economy. One farmer stated bluntly in 1996: "The government should think about us. It has plenty of land it can give us [points to the national park]"—a bold demand with an undercurrent suggesting the greed of government officials, something that would have been almost unthinkable in the political climate established under the highly authoritarian Banda regime. In addition, challenges were made against the moral authority of park rangers including accusations of violent beatings and rape of villagers. Some claims appeared to be fabricated. This new rhetoric appeared to be constructed, at least partly, with the aim of undermining the legitimacy of the government's control of park land, thereby expanding the moral basis upon which villagers could claim resources, and even land, in the park.

Whether these expanding claims have or will result in radically altered patterns of village resource use within the boundaries of the park remains an interesting question, and one that seems potentially amenable to assessment using the increased acuity of new remote sensing techniques, where patterns of culling might pinpoint exactly which resources are being taken, allowing field techniques to follow up on the ground. The pattern of resource exploitation in the park revealed by aerial photographs from 1962 and 1995 almost certainly reflects both demographic pressures (in combination with land alienation and migration) *and* particular cultural and political constraints. By the 1990s, inadequate landholding sizes had become a significant problem for many farmers in the Kasungu area, despite the relative abundance of land in the region overall. The pattern of thinning, but not wholesale clearing, of forests in the national park suggests the gradual increase of land pressure over time among these villagers.

The mid-1990s, however, brought a political watershed, with new and expanded moral claims to resources within the park boundaries. Whether these new claims actually resulted in a radical shift in resource use within the park (for example, outright clear-cutting and cultivation with the park boundaries) is a question that has not yet been examined, but would seem to be answerable through the use of recent aerial photographs or high-resolution satellite imagery.[4]

Discussion: Social Causation and Environmental Change in a Heterogeneous Temporal Landscape

In our research we turned to remote sensing (specifically, time-series aerial photography) to help answer a key question: to the extent that a deforestation 'crisis' exists, what are the key social factors that might be driving this problem? In some respects this is a question that is relatively easily addressed through remote sensing, since forests are among the landscape features that are easiest to identify through remote sensing. But there are also clear limitations. For example, we would have liked to be able to assess shifts between crops and cropping practices as small farmers attempt to diversify and intensify their production in response to economic and environmental pressures—a task that was not possible with the remote sensing technology available in the mid and late 1990s (although we understand that new technology such as high spatial resolution hyperspectral mapping may enable such analysis in the future). As social scientists with little remote sensing experience, we were at first relieved to find that aerial photography available since the 1960s for Malawi (and in a few cases the 1950s) could relatively easily identify patterns of change in forest cover, even with relatively low resolution 1:40,000-scale black and white photography. We quickly found, however, that the limitations of technology were not the biggest obstacle to making sense of the images. Bigger challenges appeared to arise from the social-temporal complexities of our research sites.

Multiple Causation at Multiple Temporal Scales

Our first observation is extremely elementary, yet may bear discussion as it did not come to our attention immediately. Although the remote sensing images could reveal important changes in forest cover, these changes occurred over periods characterized by multiple important social changes occurring simultaneously at different time scales (i.e., long-term gradual and continuous process; shorter-term abrupt changes; and 'punctuated' events that occur once at a brief point in time), making it very difficult to infer causal relationships with observed changes in forest cover. Which of these multiple processes explained the changes we saw

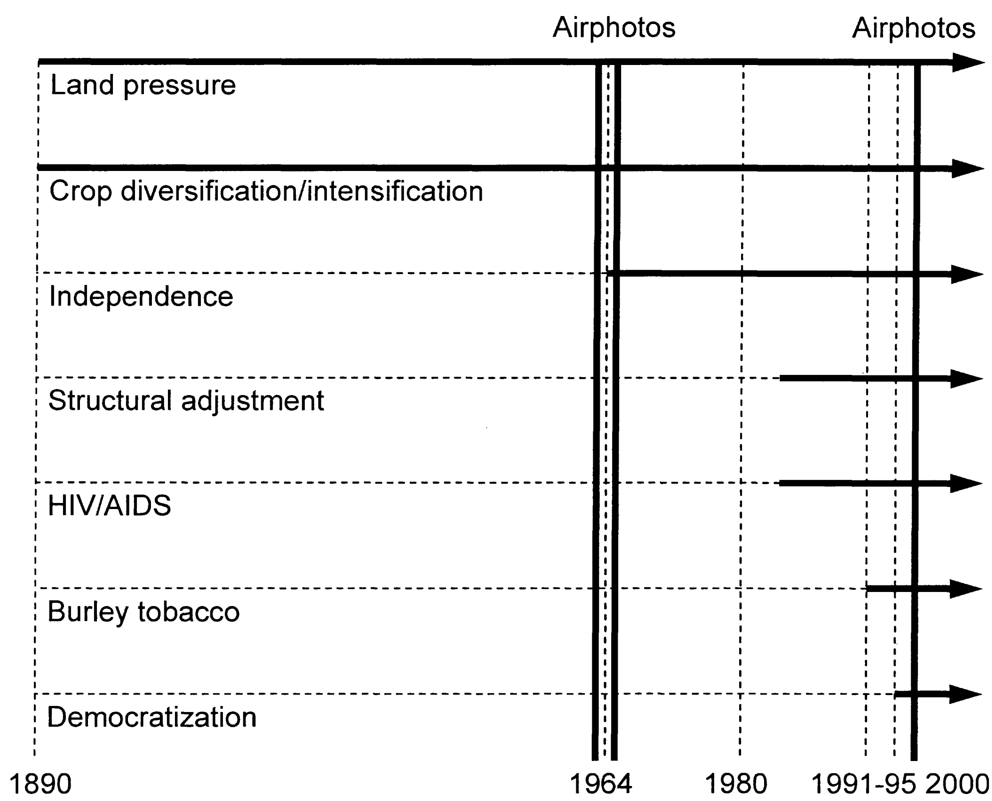
[4] In the period after 'democracy' in 1994, state capacities including land surveys and aerial photography unfortunately rapidly declined along side an explosion of petty corruption and chronic financial crisis.

'on the ground?' We had (certainly rather naïvely) believed that our on-the-ground ethnography would 'explain' the environmental changes observed from the air in our time-series photography. We quickly discovered that linking our social research to observed environmental changes was much more difficult, not least because the temporal frame of available remote sensing images did not match the temporal frame of many of the most important social changes we could document with precision. To begin, we could not pretend to have a baseline date with comparable data for all important factors. Many of the long-term changes (i.e., land pressure, crop diversification) that appear most relevant to forest change long predated even the earliest aerial photography in this region. Moreover, some of these social changes occurred at different points in time, and at different rates, not corresponding to the environmental data, making the task of parsing out social causation still more difficult. Figure 1 summarizes selected social changes occurring in our research areas since the late nineteenth century, as discussed in the previous sections.

A perhaps less obvious difficulty in assessing the effects of land pressure is that these pressures occurred simultaneously with (indeed, were very likely the primary stimulus for) efforts by small farmers to intensify and diversify their crop production rather than, or in addition to, pursuing extensive strategies such as migration and bringing more land into cultivation. Even putting aside the lack of earlier aerial photography and the technical challenges of identifying evidence of agricultural diversification from 1:40,000-scale black-and-white images, the very simultaneity of these symmetrical processes makes the task of teasing apart the presumed negative effects of land pressure from the presumably buffering effects of diversification and intensification on loss of forest cover extremely daunting. Because these critically important processes occur over a long period simultaneously, there is no 'before and after' method to gauge their impacts. To address these problems, we used the comparison of different geographic regions (Zomba and Kasungu Districts) with very different population densities to gauge the impacts of land pressure on deforestation, but found that the Boserupian effects of crop diversification and intensification in the highly populated Zomba District made it impossible to identify the effects of land or population pressure isolated from corresponding effects of diversification. The simultaneity of these interconnected processes thus contributed to the apparent near-impossibility of answering such critically important questions as how much deforestation might have occurred in Zomba District had diversification and intensification of crop production *not* occurred, or indeed to parse out how much and what kind of deforestation is due to land pressure for cultivation versus need for wood versus commercial incursions.

This problem of simultaneity was compounded by the fact that the earliest aerial photographs available to us (1962 and 1965) happened to be from precisely the period of Independence when some important political changes

Fig. 1 Summary of selected social changes occuring in our research areas since the late nineteenth century.



occurred, including land reallocation and the relaxation (at least temporarily) of some of the most severe colonial conservation policies. This precluded the intriguing possibility of making 'before and after' comparisons to assess the effects of such critical events. We can clearly see in our aerial photography that significant deforestation occurred during this time, but because we have no baseline data on the rate of deforestation prior to Independence, we saw no way to isolate the rate of deforestation associated with long-term gradual processes of increasing land pressure from the relatively abrupt changes in political economy that occurred at Independence.

Similarly, and quite regrettably for analysis, two other critically important social changes also appear to suffer from a problem of simultaneity. As illustrated in Fig. 1, the first rounds of social and economic changes induced through structural adjustment policies (SAPs) took place in the mid-1980s at almost precisely the time that the earliest effects of the catastrophic HIV/AIDS crisis began. Although in this case (unlike land pressure and crop diversification) there is no obvious causal relationship, the near-simultaneous appearance of the important social changes associated with both SAPs and AIDS almost uniformly across the region makes assessment of the independent impacts of these events on deforestation highly problematic.

The difficulties of using remote sensing in any one-to-one correspondence to assess the impacts of social processes or events that occur simultaneously or near-simultaneously with other major social events or processes drew our attention to the importance of carefully considering temporal frames in research design and to the need to move back and forth from the social to the environmental analysis. In Malawi, some of these problems could have been avoided, for example if we had searched harder for earlier aerial photographs prior to Independence. Other problems of simultaneity, such as the intertwined, long-term effects of land pressure/population growth and crop diversification may be nearly impossible to overcome or may require special methods; in either event our experiences as social scientists and novice users of remote sensing suggested the importance of early consideration of temporal frames in creating effective research designs; or at least in maintaining realistic expectations.

Finally, we found that social differentiation is highly relevant at more than one level, as exemplified in more than one institutional complex. It cannot be reduced to one variable or one indicator. The institutions that mediate resource access between social strata, or differently endowed local actors, differ from one context to another, due to the historical processes by which such endowments were established in the first place and bolstered or undermined by later legal and policy developments.

The Problem of 'Visible' and 'Invisible' Processes and Events

It has frequently been observed that certain types of social changes manifest themselves on the landscape in ways that are relatively easy to detect through remote sensing, while other processes may be either impossible to adequately detect or require special techniques such as the use of 'proxy' measures. We can only add to this important point by observing that the temporal scale at which certain social processes or events occur may be an important factor in determining their relative 'visibility' through remote sensing. For example, as discussed above, in our research area the impacts of growing pressure on land are clearly expressed on the landscape in ways that are easy to detect through fine-scale remote sensing imagery: even with 1:40,000-scale black and white aerial photographs, the roads, human-made structures (houses, buildings, barns), cleared land, cultivated hillslopes, and other features associated with increased human population density are relatively easily detected. In contrast, the highly important processes of crop diversification and intensification are much more difficult to detect and could be all too easily ignored. Crop diversification and intensification are strategies of particularly great importance to the poorest smallholders, raising the troubling question of whether remote sensing 'sees' the land uses of particular social groups and not others.

Just as certain types of changes may be difficult to detect in part because they happen over a long period in symmetry with other more 'visible' processes, certain types of changes may be easy to miss in the interpretation of remote sensing images because they happen over a shorter term or more abruptly. In our case studies the relatively abrupt and dramatic effects of SAPs and HIV/AIDS are not only difficult to 'see' because they are simultaneous but they could be easily missed because they happened so over such a short time, at least relative to long-term processes such as increased land pressure and crop diversification/intensification. One of the critical challenges of using remote sensing in Africa (especially for long-term historical work) is the infrequent periods for which images are available. The long duration of processes such as land pressure or even crop diversification and intensification (putting aside the other challenges we have discussed) makes it possible to use aerial photographs opportunistically. For more recent events satellite images may be available on a continuous temporal basis, but it then becomes incumbent upon social scientists to identify the 'right' moments to observe—for example, the sudden importance of the Madsen estate that we described (a shift that was particularly important for social groups with particular kinds of historical claims on estate land).

The Challenge of 'Punctuated' Social Events

Another challenge associated with temporal framing that we experienced draws attention to the importance of 'punctuated' social events—those processes which may leave a significant mark on the landscape, but appear abruptly and dissipate quite quickly. In our case studies, this problem arose in interpreting the effects of Malawi's experience of democratization that came into full force in 1994 (Fig. 1)—an event that was particularly potent because of the country's long history of extreme political oppression and associated social differentiation and deprivation. Democracy in 1993–1994 was interpreted by some politicians and the general public, especially the poor, as an undoing of all previously binding restrictions. This attitude was promoted by politicians who, in apparent efforts to pander for votes, hinted that previously protected natural resources would be made into public goods. As a consequence, in 1994, citizens took over numerous forested areas (both state and private) and engaged in clear-cutting at incredible rates, suspending the institutional barriers between the resources of the state, the rich and the rest. Whole forests in periurban areas disappeared almost literally overnight (Walker, 1999; press and other reports continue to indicate forest clearing over most of the country continue today, at a less dramatic pace).

Because of the highly punctuated nature of these historical events, and despite the almost certain enormous impacts on some local landscapes, the impacts of 'democracy' on the landscape are all but invisible when one looks at longer time periods through which other processes are highlighted: for example, the introduction and rapid increase in burley tobacco production (which requires large amounts of tree products) by small farmers that occurred at almost the same time (Fig. 1). Although we have relatively high quality (1:25,000-scale) aerial photographs for our research sites from 1995, these were taken *after* the peak period of deforestation resulting from the 'democracy' free-for-all. Assessment of the impacts of this relatively brief but intense historical event would require tight temporal bracketing of remote sensing images, with, say, paired aerial photography from 1993 and 1995. As already mentioned, in Malawi aerial photography is available infrequently and irregularly, making this sort of tight temporal bracketing virtually impossible except by the use of satellite images—although the resolution of satellite data at that time may not have allowed for detailed analysis at the relevant geographic scales. In sum, our experiences suggest that social scientists who are not experts in remote sensing need to be aware of both the importance of temporal scale (in this case, the need for tight temporal bracketing) and also of the types of emerging technologies that may make these previously largely invisible 'punctuated' events more visible.

Conclusions

Our experiences as social scientists and novice users of remote sensing and GIS in Malawi have suggested to us that there is no easy way through remote sensing images alone to detect whether the increased clearing and cultivation of forested land is a result of long-term ongoing land pressures and consequent efforts at crop diversification and intensification, economic pressures for cash crop production from structural adjustment policies, the expansion of smallholder burley tobacco production, the changing moral economy of land use and deforestation following 'democratization,' or other social forces. What remote sensing and GIS can tell us is that clearing of trees is an important phenomenon, and they can offer precision on the locations and the types of clearing and forest use taking place. Our expectations of what remote sensing and GIS can tell us about the social driving forces of these changes should be tempered, not least because of the issues of temporal framing and simultaneity that we have discussed.

Even as remote sensing becomes ever more available and sophisticated, social scientists and remote sensing experts alike would do well to keep in mind that some kinds of social and environmental processes (especially those occurring at certain types of temporal and geographic scales) may simply never be 'visible.' For example, the things that we 'see' in remotely sensed images are not necessarily what people perceive on the ground. This is, of course, the subject of much of the critical literature on remote sensing. In our social research on deforestation we found that although time-series aerial photography revealed unambiguous trends toward deforestation, people on the ground did not necessarily perceive a 'crisis.' On the other hand, the perspective of the remote sensing scientist is in some ways parallel to the perspective of government administrators and scientific advisors who have, since the colonial era, perceived a 'crisis' of deforestation in Malawi. The difference, we suggest, is that the view-from-the-sky offered by long-term analysis at very broad temporal and geographic scales hides the on-the-ground realities of what we might call 'niche' economies (or ecologies). As we discussed, many respondents in our research sites perceive that there is no absolute scarcity of trees; there is only a problem of access to pockets of forests that remain. In this case there is no one 'right' scale. Our experiences suggest the importance for both social scientists and remote sensing experts of understanding that differences in temporal scale may reveal different truths to different people—and that these differences are highly influenced by particular histories and forms of social stratification.

We hope that our (no doubt naïve) experiences described in this article may stimulate increased discussion about the challenges and opportunities that a greater focus on temporal scale and social differentiation offers for understanding social and environmental change through integrated social science and remote sensing research.

References

Benson, T., Kaphuka, J., Chinula, R., and Kanyanda, S. (2002). *Malawi: An Atlas of Social Statistics*. Malawi National Statistical Office, Zomba, Malawi and the International Food Policy Research Institute, Washington, DC.

Craig, W. J., Harris, T. M., and Weiner, D. (2002). *Community Participation and Geographic Information Systems*. Taylor & Francis, London, New York.

DeSanker, P. V., Justice, C. O., Munthali, G., and Masamvu, K. (2005). Requirements for integrated assessment modelling at the regional and national levels in Africa to address climate change. In Sum, P. (ed.), *Climate Change and Africa*. Cambridge University Press, Cambridge.

Dickerman, C. W., and Bloch, P. C. (1991). *Land Tenure and Agricultural Productivity in Malawi*. Land Tenure Center, Madison, Wisconsin.

Fox, J. (2003). *People and the Environment: Approaches for Linking Household and Community Surveys to Remote Sensing and GIS*. Kluwer, Boston.

Geoghegan, J., Villar, S. C., Klepeis, P., Mendoza, P. M., Ogneva-Himmelberger, Y., Chowdhury, R. R., Turner, B. L., and Vance, C. (2001). Modeling Tropical Deforestation in the Southern Yucatan Peninsular Region: Comparing Survey and Satellite Data. *Agriculture, Ecosystems & Environment* 85 (1–3): 25–46.

Geoghegan, J. M., and Gray, W. B. (2005). *Spatial Aspects of Environmental Policy, The International Library of Environmental Economics and Policy*. Ashgate, Aldershot, Hampshire, England.

Heasley, L. (2003). Shifting Boundaries on a Wisconsin Landscape: Can GIS Help Historians Tell a Complicated Story? *Human Ecology* 31 (2): 183–213.

Jiang, H. (2003). Stories Remote Sensing Images Can Tell: Integrating Remote Sensing Analysis with Ethnographic Research in the Study of Cultural Landscapes. *Human Ecology* 31 (2): 215–232.

Kambewa, B. M. D. (2006). *Access and Control in Lake Chilwa Wetlands: An Analysis of Social and Power Relations*, Ph.D. dissertation, Bunda College of Agriculture, University of Malawi, Bunda.

Kydd, J., and Christiansen, R. E. (1982). Structural Change in Malawi Since Independence: Consequences of a Development Strategy Based on Large Scale Agriculture. *World Development* 10 (5): 355–375.

McCracken, J. (1987). Colonialism, capitalism and the ecological crisis in Malawi: a reassessment. In Anderson, D., and Grove, R. (eds), *Conservation in Africa: people, policies, and practice*. Cambridge University Press, Cambridge, pp. 63–77.

Mhone, G. C. Z. (1987). Agricultural and food policy in Malawi: a review. in Mkandawire, T., and Bourenane, N (eds.), *The State and Agriculture in Africa*. CODESRIA, Dakar, Senegal.

Mwase, G. S., and Rotberg, R. I. (1967). *Strike a Blow and Die; A Narrative of Race Relations in Colonial Africa*. Harvard University Press, Cambridge, MA.

Orr, A., and Mwale, B. (2001). Adapting to Adjustment: Smallholder Livelihood Strategies in Southern Malawi. *World Development* 29 (8): 1325–1343.

Peters, P. E. (2002). The limits of knowledge: securing rural livelihoods in a situation of resource scarcity. In Barrett, C. B., Place, F., and Aboud, A. A. (eds.), *Contested Resources: Challenges to the Governance of Natural Resources in Southern Africa*. CABI, London.

Robbins, P. (2003). Beyond Ground Truth: GIS and the Environmental Knowledge of Herders, Professional Foresters, and other Traditional Communities. *Human Ecology* 31 (2): 233–253.

Sheppard, E. S., and McMaster, R. B. (2004). *Scale and Geographic Inquiry: Nature, Society, and Method*. Blackwell, Malden, MA.

Thompson, E. P. (1975). *Whigs and Hunters: The Origin of the Black Act*. Allen Lane, London.

Turner, M. D. (2003). Methodological Reflections on the Use of Remote Sensing and Geographic Information Science in Human Ecological Research. *Human Ecology* 31 (2): 255–279.

Turner, M. D., and Taylor, P. J. (2003). Critical Reflections on the Use of Remote Sensing and GIS Technologies in Human Ecological Research. *Human Ecology* 31 (2): 177–182.

Turner, B. L., Geoghegan, J. M., and Foster, D. R. (2004). *Integrated Land-change Science and Tropical Deforestation in The Southern Yucatán: Final Frontiers, Clarendon Lectures in Geography and Environmental Studies*. Oxford University Press, Oxford, New York.

United Nations. (1993). *Malawi Situation Analysis of Poverty*. Lilongwe: United Nations in Malawi.

United Nations Department of Economic and Social Affairs/Population Division. (2003). *The Impact of AIDS*. United Nations, New York.

Utrecht University Library. (2005a). *Malawi historical demographical data of the administrative division*. Malawi National Statistical Office 2002 [cited April 6 2005]. Available from http://www.library.uu.nl/wesp/populstat/Africa/malawip.htm.

Utrecht University Library. (2005b). *Malawi: historical demographical data of the whole country*. Malawi National Statistical Office 2002 [cited April 6 2005]. Available from http://www.library.uu.nl/wesp/populstat/Africa/malawic.htm.

Vail, L. (1975). The Making of an Imperial Slum: Nyasaland and its Railways, 1895–1935. *Journal of African History* 16 (1): 89–112.

Vail, L. (1983). The political economy of East-Central Africa. In Birmingham, D., and Martin, P. M. (eds.), *History of Central Africa, Vol. 2*. Longman, London.

Vance, C., and Geoghegan, J. (2002). Temporal and Spatial Modelling of Tropical Deforestation: A Survival Analysis Linking Satellite and Household Survey Data. *Agricultural Economics* 27 (3): 317–332.

Walker, P. (1999). Democracy and Environment: Congruencies and Contradictions in Southern Africa. *Political Geography* 18 (3): 257–284.

Walker, P. A. (2004). Roots of Crisis: Historical Narratives of Tree Planting in Malawi. *Historical Geography* 32: 89–109.

Walker, P. A., and Peters, P. E. (2001). Maps, Metaphors, and Meanings: Boundary Struggles and Village Forest use on Private and State Land in Malawi. *Society & Natural Resources* 14 (5): 411–424.

White, L. (1985). *Magomero: Portrait of An African Village*. Cambridge University Press, Cambridge.

Zubrow, A. A. S. (2003). Mapping Tension: Remote Sensing and The Productian of a Statewide Land Cover Map. *Human Ecology* 31 (2): 281–307.